

Advanced (Carolina Reaper)

Q1. What is the expansion of $(x + 3)^2$?

- (A) $x^2 + 6x + 9$
 (B) $x^2 + 3x - 9$
 (C) $x^2 - 6x + 9$
 (D) $x^2 + 9x + 3$
 (E) $x^2 - 9x + 3$

Q2. Which expression is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
 (B) $x^2 + 4x - 4$
 (C) $x^2 - 4x + 4$
 (D) $x^2 + 4x + 5$
 (E) $x^2 - 4x - 4$

Q3. A bookshelf has x shelves, and each shelf can hold $x + 2$ books. What is the total number of books the bookshelf can hold?

- (A) $x(x + 2)$
 (B) $x(x - 2)$
 (C) $x^2 + 2x$
 (D) $x^2 - 2x$
 (E) $x^2 + 4$

Q4. Factor the expression $x^2 + 10x + 25$

- (A) $(x + 5)^2$
 (B) $(x - 5)^2$
 (C) $(x + 5)(x - 5)$
 (D) $x(x + 10) + 25$
 (E) $x(x - 10) - 25$

Q5. Tom has been saving money for a new bike and has x dol-lars in his piggy bank. He wants to buy a bike that costs $x^2 + 5x + 6$ dollars. What is the expansion of the expression?

- (A) $(x + 3)(x + 2)$
 (B) $(x - 3)(x + 2)$
 (C) $(x + 3)(x - 2)$
 (D) $x^2 + 5x - 6$
 (E) $x^2 - 5x + 6$

Q6. A rectangular garden has a length of $x + 3$ meters and a width of $x - 2$ meters. What is the area of the garden?

- (A) $x^2 + x - 6$
 (B) $x^2 + 5x + 6$
 (C) $x^2 - x - 6$
 (D) $x^2 + x + 6$
 (E) $x^2 - 5x - 6$

Q7. What is the factored form of $x^2 - 7x + 12$?

- (A) $(x - 3)(x - 4)$
 (B) $(x + 3)(x + 4)$
 (C) $(x - 3)(x + 4)$
 (D) $(x + 3)(x - 4)$
 (E) $x(x - 7) + 12$

Q8. Ashley is making a quilt with x rows of squares, and each row has $x + 2$ squares. What is the total number of squares in the quilt?

- (A) $x(x + 2)$
 (B) $x(x - 2)$
 (C) $x^2 + 2x$
 (D) $x^2 - 2x$
 (E) $x^2 + 4$

Open Ended Questions (FRQ)**Q9.** A water tank can hold $x^2 + 5x + 6$ gallons of water. Factor the expression and explain its meaning in the context of the water tank.**Q10.** A bakery is making a special batch of $x^2 + 10x + 25$ pastries. Factor the expression and describe how the bakery can use this information to package the pastries.**Chef's Tips**

- Provide clear instructions for each question
- Emphasize the importance of factoring and expanding special patterns
- Encourage students to use real-life examples to illustrate their answers